

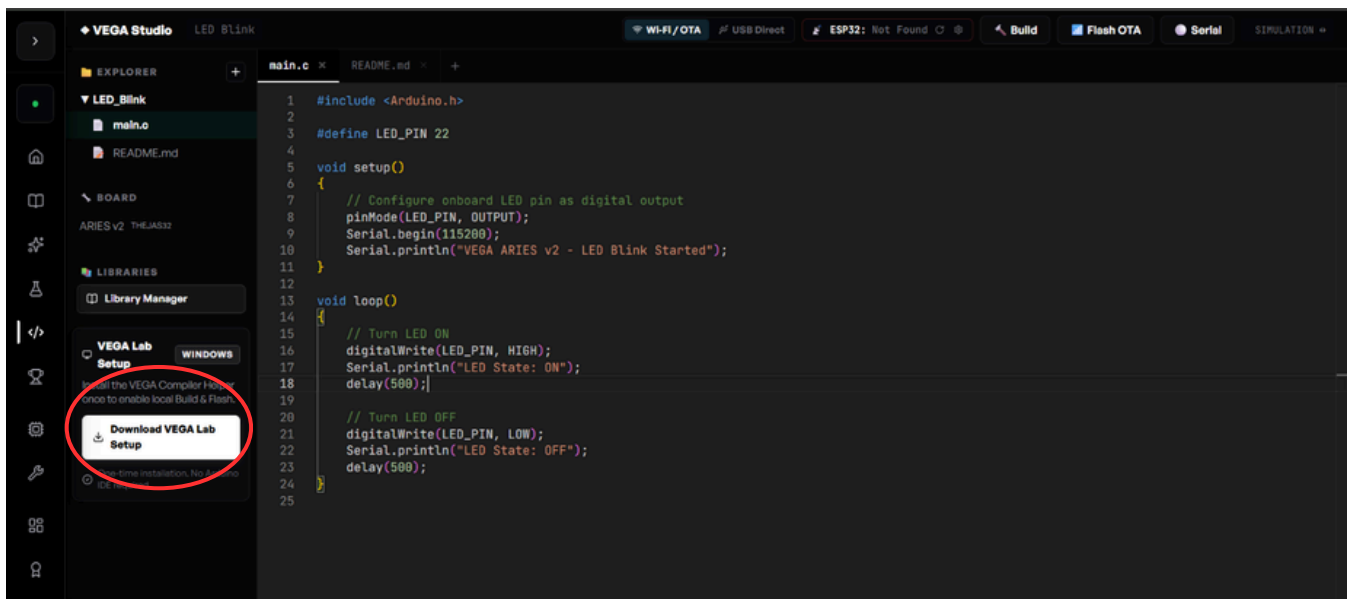
VEGA EDULINK — VEGA Lab Setup & Build Manual

Step-by-step procedure to install VEGALab, start the compiler service, open the WebIDE, and generate the VEGA firmware.bin file.

Prerequisite: A Windows PC and access to the VEGA EDULINK Web IDE.

Step 1 — Open the VEGA IDE website

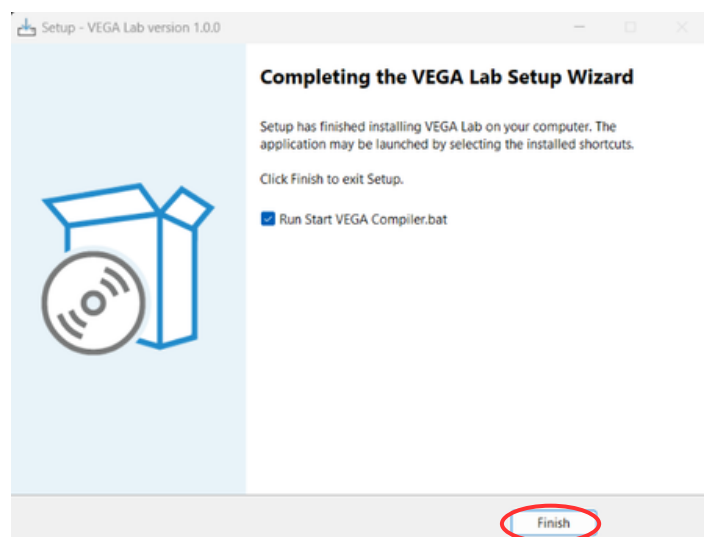
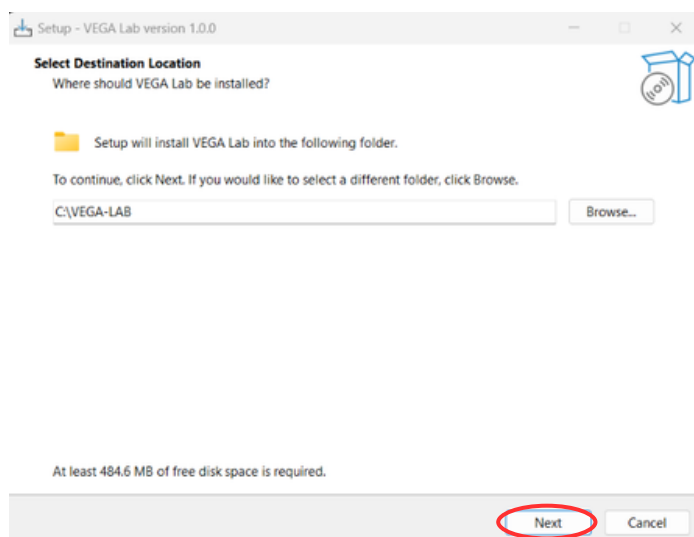
Open the VEGA EDULINK Web IDE in your browser. On the home page, click Download VEGA Lab Setup.



Download the VEGA Lab Setup installer

Step 2 — Install VEGA Lab Setup

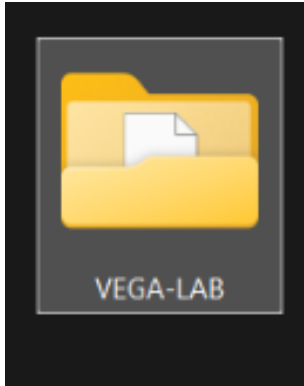
Run the downloaded installer. **Do not change the installation path.** Keep the default location on Local Disk (C:).



Expected location: C:\VEGA-LAB

Step 3 — Start the VEGA Compiler

Open File Explorer and go to **Local Disk (C:) → VEGA-LAB → service**. Locate **Start VEGA Compiler** and run it.



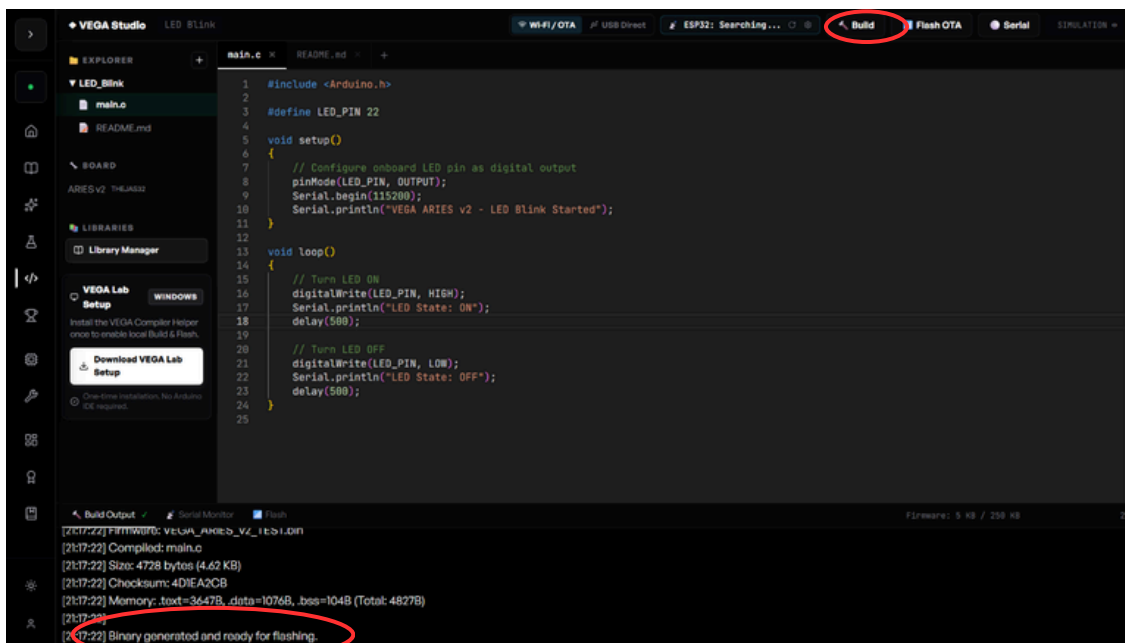
Name	Date modified	Type	Size
compiler	27-09-2026 09:22 PM	File folder	
core	27-09-2026 09:22 PM	File folder	
hardware	27-09-2026 09:22 PM	File folder	
libraries	27-09-2026 09:22 PM	File folder	
runtime	27-09-2026 09:22 PM	File folder	
service	27-09-2026 09:22 PM	File folder	
unins000.dat	27-09-2026 09:22 PM	DAT File	1,508 KB
unins000	27-09-2026 09:21 PM	Application	4,349 KB
VEGA-Lab-Setup	07-09-2026 10:05 PM	Inno Setup Script	1 KB

Name	Date modified	Type	Size
node_modules	27-09-2026 09:22 PM	File folder	
package	07-09-2026 04:46 PM	JSON Source File	1 KB
package-lock	07-09-2026 04:46 PM	JSON Source File	31 KB
server.backup	07-09-2026 03:27 PM	JavaScript Source ...	14 KB
server	09-09-2026 12:50 AM	JavaScript Source ...	27 KB
Start VEGA Compiler	07-09-2026 09:41 PM	Windows Batch File	1 KB

Keep the compiler service running while using the Web IDE.

Step 4 — Open the VEGA IDE and click Build

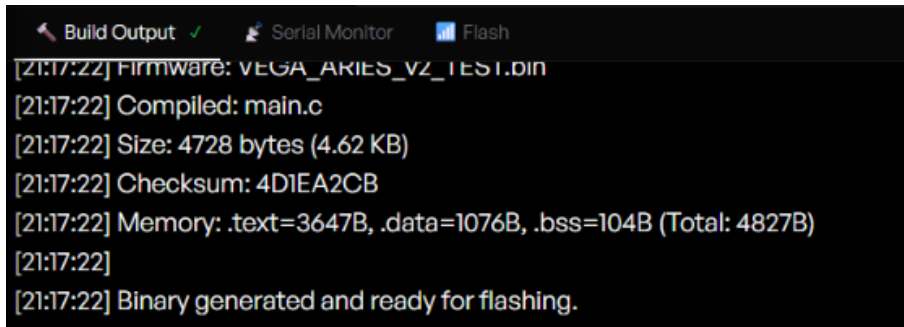
Open the VEGA EDULINK WebIDE ,write or open your source code,and click **Build**.



The IDE sends the source files to the local compiler service, which builds the firmware for the VEGA target.

Step 5 — Generate the .bin firmware

After a successful build, the required **.bin** firmware file is generated.



```
Build Output ✓ Serial Monitor Flash
[21:17:22] Firmware: VEGA_ARIES_V2_TEST.BIN
[21:17:22] Compiled: main.c
[21:17:22] Size: 4728 bytes (4.62 KB)
[21:17:22] Checksum: 4D1EA2CB
[21:17:22] Memory: .text=3647B, .data=1076B, .bss=104B (Total: 4827B)
[21:17:22]
[21:17:22] Binary generated and ready for flashing.
```

Steps for OTA Setup:

Step 1: Power the ESP32-S3.

Step 2: Connect the ESP32-S3 to the required Wi-Fi network by first dumping the Wi-Fi firmware/code into the ESP32-S3. Then connect the laptop/PC to the **same Wi-Fi network**.

Step 3: The ESP32-S3 has two USB ports. Use the USB port required for communication with the VEGA board.

Step 4: Connect the ESP32-S3 USB port to the Type-C USB cable.

Step 5: Connect the other end of the USB cable to the OTG cable, and connect the OTG cable to the VEGA ARIES v2 board.

Step 6: Open the VEGA Web IDE. Once the ESP32-S3 and laptop/PC are connected to the same network, the ESP32-S3 will be automatically discovered and its IP address will be shown in the Web IDE. This confirms that the ESP32-S3 is connected.

Step 7: Open or write the required source code in the VEGA Web IDE and click Build. The source code is compiled and the .bin firmware is generated.

Step 8: Start the OTA Flash process. The generated .bin file is sent from the Web IDE to the ESP32-S3 through Wi-Fi. The ESP32-S3 then transfers the firmware to the VEGA ARIES v2 through the USB OTG connection.

Step 9: Once the Web IDE shows “Flashed Successfully”, press the RESET button on the VEGA ARIES v2 board.

Step 10: Wait for a few seconds. The firmware will be programmed into the VEGA board and the uploaded application will start running.

Make sure the wifi is in **2.4Ghz and WPA2 Personal**